

# The Application Layer: DNS

The Internet's Phonebook: How We Find Websites

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# Recap: The HTTP Conversation

## Last Week: HTTP

We learned that your browser (Client) talks to a Server using **GET** requests.

### The Problem:

- Humans speak **Names**: `www.google.com`
- Computers speak **Numbers**: `142.250.189.68` (IP Address)

## The Missing Link

Before your browser can send that `GET /index.html` request, it needs to find the Server's IP address first.

**Today's Topic:** How that translation happens.

# DNS: The Domain Name System

## Definition

**DNS** is a distributed database that translates **Hostnames** into **IP Addresses**.

## The Analogy: The Phonebook

- If you want to call "Pizza Hut", you look up the name to find the number.
- If you want to visit "Concordia", DNS looks up the name to find the IP.

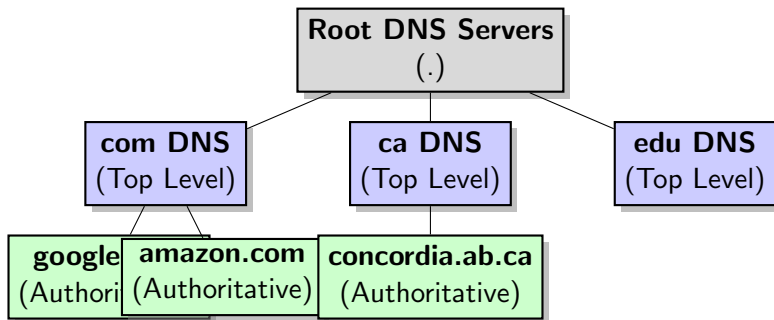
## Why "Distributed"?

- No single computer has all the names (Too big! Too slow!).
- Instead, the list is split up across thousands of servers.

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Source: Kurose & Ross: Section 2.4.1 (Services Provided by DNS)

# The DNS Hierarchy



## The Query Path:

- 1 **Root:** "I don't know concordia.ab.ca, ask the **.ca** server."
- 2 **TLD (.ca):** "I don't know the IP, ask the **concordia** server."
- 3 **Authoritative:** "I know! The IP is 199.175.52.12."

Source: Kurose & Ross: Figure 2.19 (DNS Hierarchy)

# What's inside the Database? (Records)

It's not just IP addresses. DNS stores different "Types" of records.

| Type         | Stands For            | Meaning                                                                |
|--------------|-----------------------|------------------------------------------------------------------------|
| <b>A</b>     | <b>Address</b>        | <b>Hostname</b> → <b>IP Address</b><br>(google.com → 142.250.x.x)      |
| <b>CNAME</b> | <b>Canonical Name</b> | <b>Nickname</b> → <b>Real Name</b><br>(www.site.com → server1.aws.com) |
| <b>MX</b>    | <b>Mail Exchange</b>  | <b>Domain</b> → <b>Email Server</b><br>(@gmail.com → smtp.google.com)  |

*\*Note: This is why "A" records are for websites, and "MX" records are for email.*

Source: Kurose & Ross: Section 2.4.2 (DNS Records)

## Summary: What you need for the Lab

- 1 **DNS:** The system that maps Names to Numbers (IPs).
- 2 **Hierarchy:** Root → TLD (.com) → Authoritative (Specific Company).
- 3 **Caching:** Your computer remembers these answers so it doesn't have to ask every time.
- 4 **The Tool:** nslookup

### Preview: Using nslookup

In the upcoming lab, we will use this tool to query the phonebook manually:

```
> nslookup google.com  
Address: 142.250.189.68
```

**Next Class:** We will start writing Python code to use these Sockets!